

Tarea dirección 9.7

1294
399

3) Máx. absoluto = 3

Min. absoluto = r Nada = a, d

Máx. locales = c

Min. locales = r, b

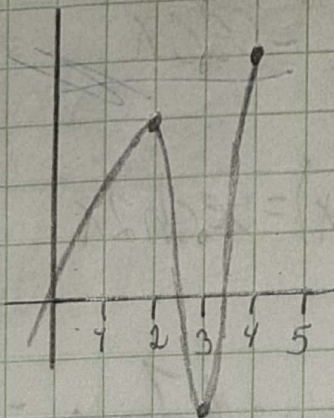
5) Máx. absoluto = $f(4) = 5$

Min. absoluto = $f(2) = 2$

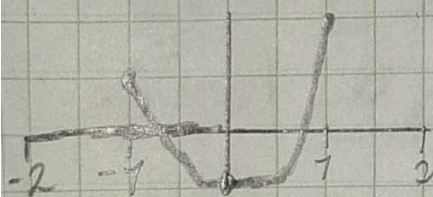
Máx. local = $f(6) = 4$ $f(4) = 5$

Min. local = $f(5) = 3$ $f(2) = 2$ $f(1) = 3$

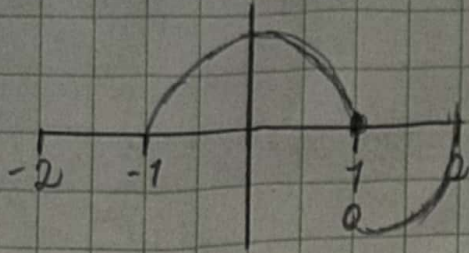
9) $D = [1, 5]$ Min. abs = 3 Máx. abs = 4 Máx. local = 2



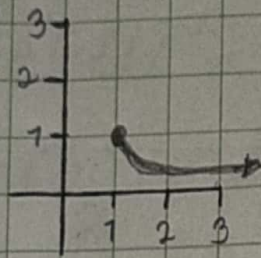
13) a) $[-1, 2]$ máx. abs, pero no min. abs.



b) $[-1, 2]$ discontinua, si mäs y min. absolute

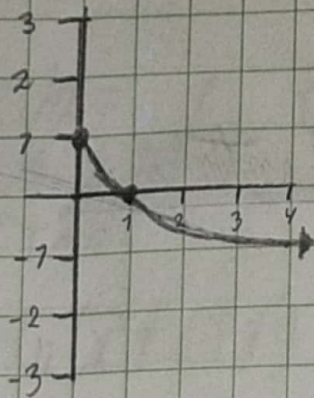


17) $f(x) = \frac{1}{x} \quad x \geq 1$



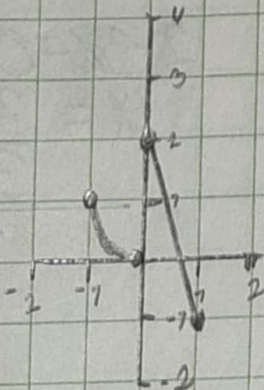
Mäs. abs. = $f(1) = 1$

25) $f(x) = 1 - \sqrt{x}$



Mäs. abs. = $f(0) = 1$

27) $f(x) = \begin{cases} x^2 & \text{si } -1 \leq x \leq 0 \\ 2-3x & \text{si } 0 < x \leq 1 \end{cases}$



Min. abs. = $f(1) = -1$

$$33) g(t) = t^4 + t^3 + t^2 + 1$$

$$g'(t) = 4t^3 + 3t^2 + t$$

$$4t^3 + 3t^2 + t = 0$$

$$t(4t^2 + 3t + 1) = 0$$

$$t = 0 \text{ número crítico}$$

$$41) f(\theta) = 2 \cos \theta + \sin^2 \theta$$

$$f'(\theta) = 2(-\sin \theta) + 2 \sin \theta \cos \theta$$

$$f'(\theta) = -2 \sin \theta + 2 \sin \theta \cos \theta$$

$$-2 \sin \theta + 2 \sin \theta \cos \theta = 0$$

$$-2 \sin \theta (1 - \cos \theta) = 0$$

$$\sin \theta = 0$$

$$\theta = 0$$

$$1 - \cos \theta = 0$$

$$\theta = 0$$

no enteros

$$43) f(x) = x^2 e^{-3x}$$

$$f'(x) = 2x e^{-3x} + x^2 e^{-3x} \cdot (-3)$$

$$f'(x) = 2x e^{-3x} - 3x^2 e^{-3x}$$

$$2x e^{-3x} - 3x^2 e^{-3x} = 0$$

$$x e^{-3x} (2 - 3x) = 0$$

$$x = 0 \quad e^{-3x} = 0$$

$$2 - 3x = 0$$

$$x = \frac{2}{3}$$

$$51) f(x) = 3x^4 - 4x^3 - 12x^2 + 1 \quad [-2, 3]$$

$$f'(x) = 12x^3 - 12x^2 - 24x$$

$$12x^3 - 12x^2 - 24x = 0$$

$$12x(x^2 - x - 2) = 0$$

$$12x(x+1)(x-2) = 0$$

$$x = 0$$

$$x+1 = 0$$

$$x = -1$$

$$x-2 = 0$$

$$x = 2$$

Scribe

$$\begin{aligned} f(-2) &= 329 \text{ Max. abs} \\ f(-1) &= 33 \\ f(0) &= 7 \\ f(2) &= -327 \text{ Min. abs} \\ f(3) &= -977 \end{aligned}$$

$$57) f(t) = t + \cot t/2 \quad [\pi/4, 7\pi/4]$$

$$f'(t) = 1 + (-\csc^2 t/2) \cdot 1/2$$

$$f'(t) = \frac{1 - \csc^2 t/2}{2}$$

$$\frac{1 - \csc^2 t/2}{2} = 0$$

$$\begin{aligned} \csc^2 t/2 &= 1 & f(\pi/6) &= \frac{3}{2}\sqrt{3} \text{ Max. abs} \\ \csc t/2 &= \pm 1 & f(\pi/2) &= 0 \text{ Min. abs} \\ t &= \pi + 2n\pi \end{aligned}$$

$$61) f(x) = \ln(x^2 + x + 1) \quad [-1, 7]$$

$$f'(x) = \frac{1}{x^2 + x + 1} \cdot 2x + 1$$

$$f'(x) = \frac{2x + 1}{x^2 + x + 1}$$

$$\frac{2x + 1}{x^2 + x + 1} = 0$$

$$\begin{aligned} 2x + 1 &= 0 \\ x &= -\frac{1}{2} \end{aligned}$$

$$\text{Max. abs} = f(1) = \ln(3)$$

$$\text{Min. abs} = f(-1/2) = \ln(3/4)$$